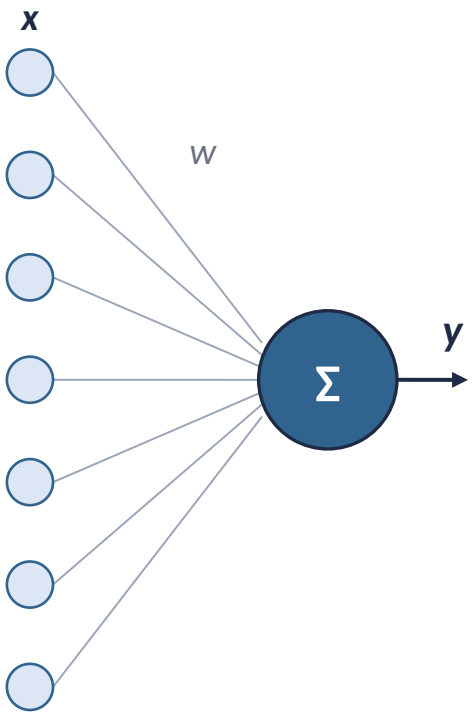
Ling 285
Human Language and Technology
Lecture 18

Neuron → NNs → LLMs → Agents!!!!

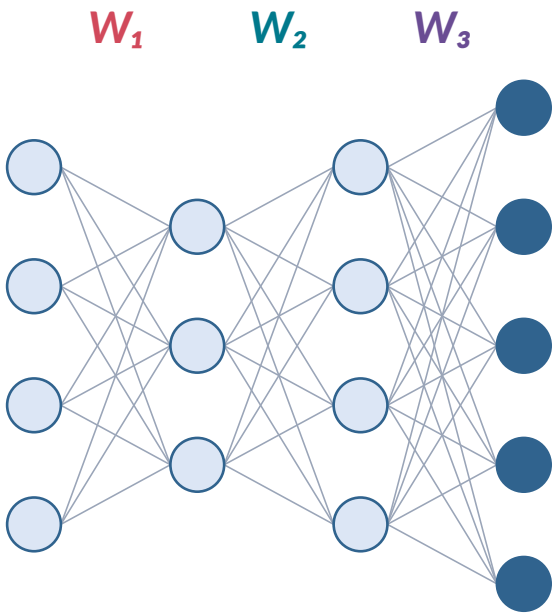
Khalil Iskarous

Department of Linguistics

University of Southern California



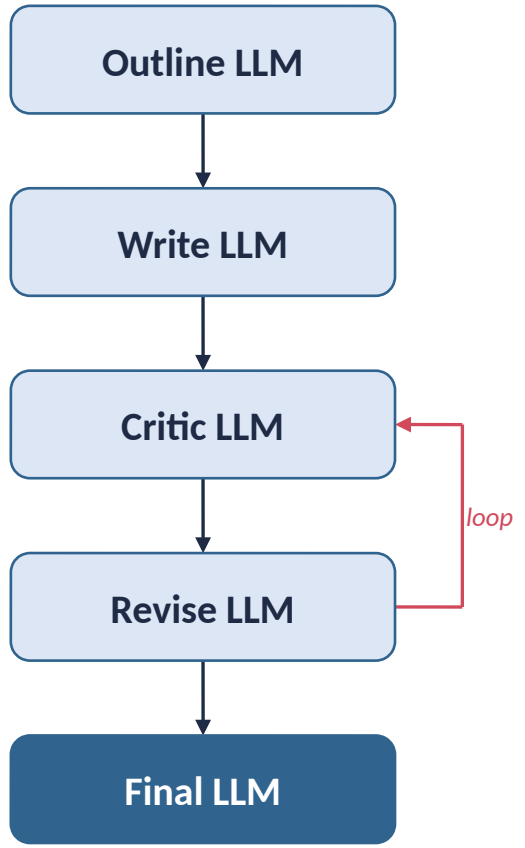
Vector in,
1 number out



Vector in,
Decision vector out



LLM: dynamical system with
many vectors in, **1 decision vector out**

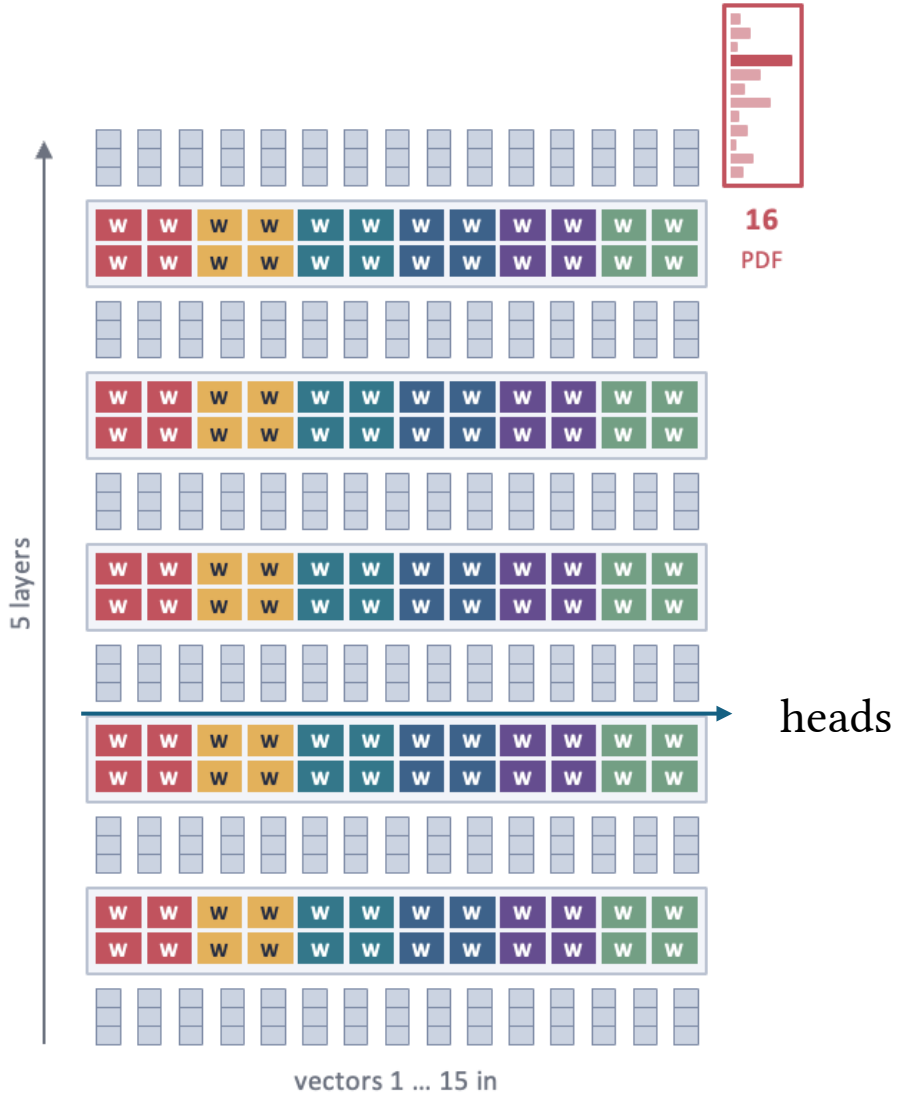


Agentic system:
a graph of LLMs

Inference: inner products
Learning: SGD + DP

The agents (LLMs) speak to each other in English, but they are multilingual in human and computer languages!!!!!!!

An LLM's task, the prediction of the next word is difficult. So difficult that it need many many *layers* of its dynamical system. But it also needs, in each layer, many pallel *heads*, where each head is a different expert, and in each layer, all the knowledge from different heads/experts is brought together to generate improved representations x. Then the same things is done yet again in the next layer.



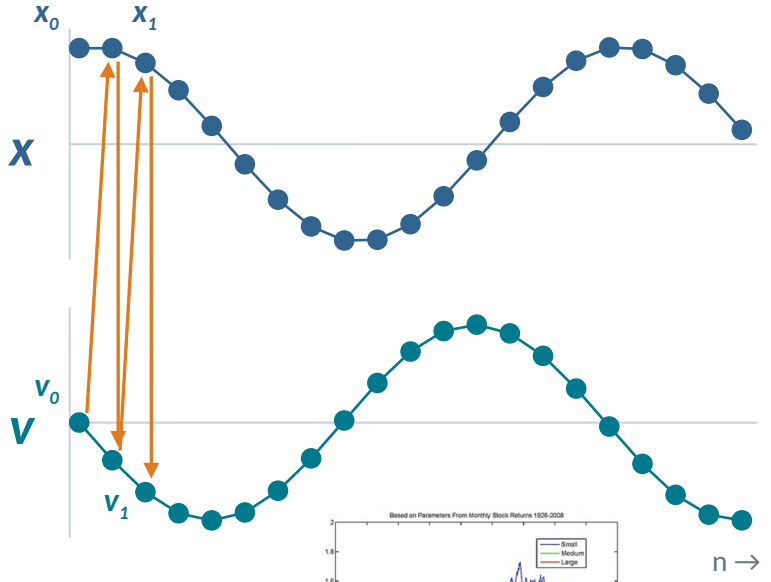
LLM: dynamical system with
many vectors in, **1 decision vector out**

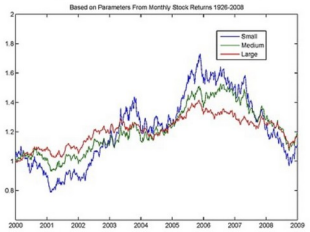
Dynamical System: A rule for data, fed back to itself



Harmonic oscillator

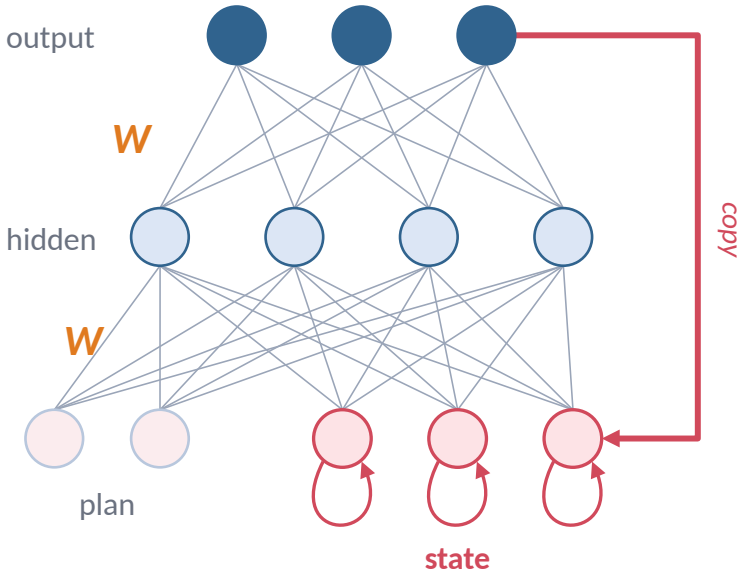
$$x_{n+1} = x_n + h \cdot v_n \qquad v_{n+1} = v_n - h \cdot x_{n+1}$$





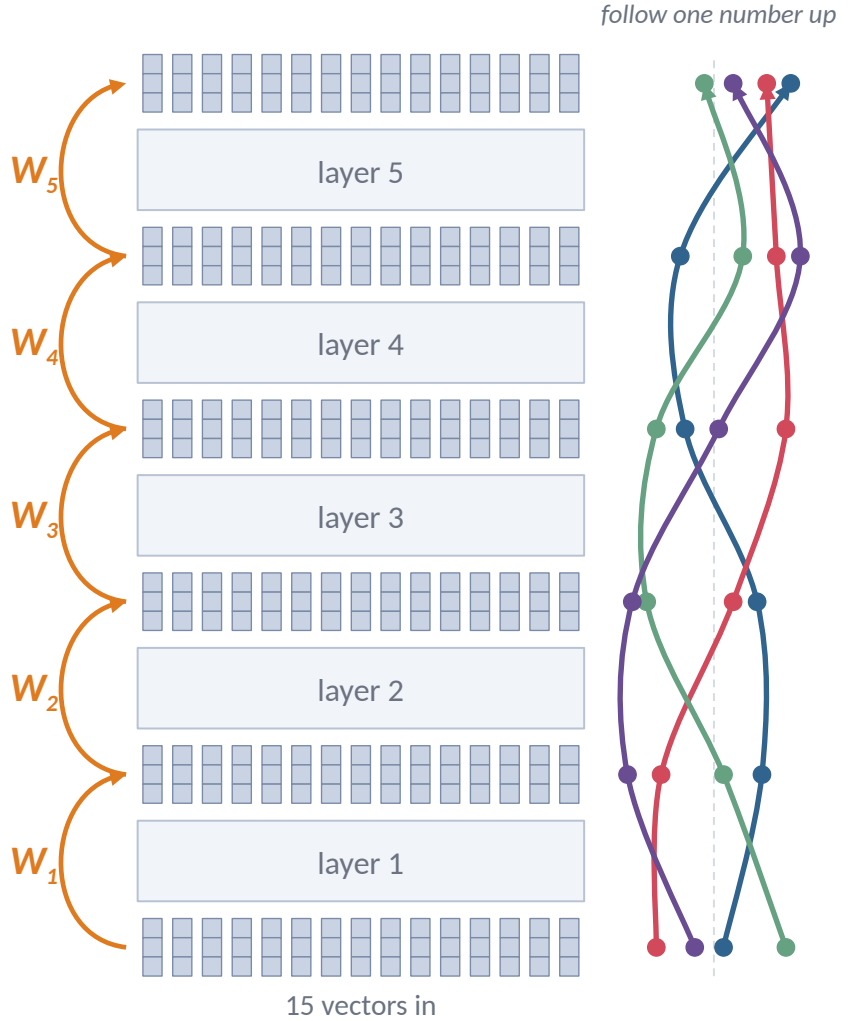
x and v take turns. Step ; a cosine.

Jordan network (1986)



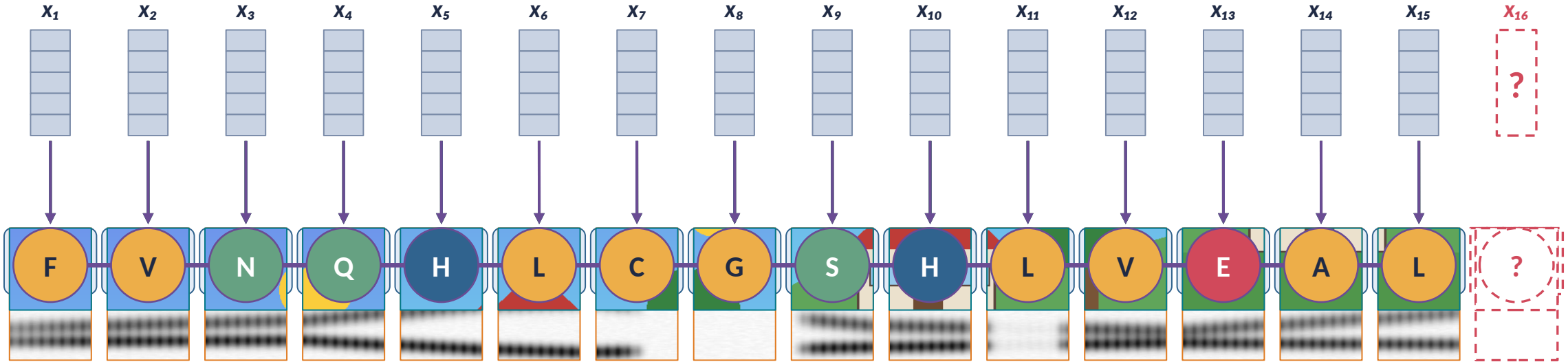
Based on time variation in speech production, Jordan proposed a marriage of NNs and Dynamical sysyems that after 30+ years of tinkering became the LLM ☺

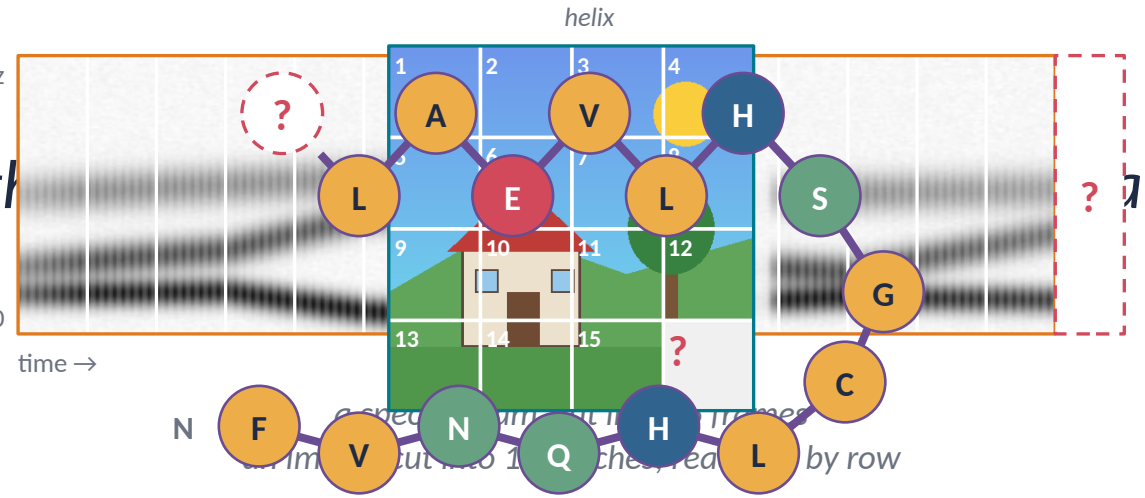
LLM: the layer is the rule



Each layer applies its W's to every vector. Depth plays the role of time: one number, followed up the layers, is a curve, just not a cosine.

15 vectors. *Vectors of what?*



The cat that th  *t on the ____*

a protein (insulin, B chain), read as 15 amino acids

The LLM only ever sees vectors. Words, sounds, pixels, molecules: one machine, one question: what is x_{16}?

Which sequences can be English words?

1 syllable, single C V C

Don	mat	sit	shave	cup	dog	pen	fish	bed	ship
kite	rope	bath	chin	wet	gum	tub	ring	lick	van

A simple syllable in English can have a consonant then a vowel then a consonant

1 syllable, clusters

onsets →					*codas →*				
split	flat	black	glad	stop	desk	box	help	milk	lamp
*sblit	*lfat	*bnack	*dlad	*tsop	*desb	*bokp	*hepl	*mikl	*lapm

Only certain sequences are allowed before the vowel and after the vowel

many syllables

ba·na·na	win·dow	pump·kin	nap·kin	At·lan·tic	ath·lete	ad·mit	spa·ghet·ti
*bnanana			*ngapkin	*tlantic	*thlete	*dmit	*sbaghetti

A long word is a chain of syllables: t·l is fine across a break (At·lan·tic), but tl cannot start a syllable

· syllable break * impossible in English

Pop Quiz!

How phonotactic knowledge can be revealed through GPT learning

Each is the start of an English word. Guess the next letter.

1

ra _

t n r l

after a vowel, almost anything

2

sp _

l r a e i o

l, r or a vowel never ***spn**

3

k _

i e a n

a vowel, or n (knee) never ***kt**

4

skr _

a e i u

only a vowel never ***skrl**

5

m _

a e i o u

only a vowel never ***mb**

6

bram _

b

bramble, or a new syllable

7

b _

r l a e i o u

r, l or a vowel never ***bn**

8

f _

r l a e i o u

r, l or a vowel never ***fn**

9

sh _

a e i o u r

a vowel, or r (shrimp) never ***shb**

Guessing the next letter is predicting x_{16}. A GPT trained only on this game learns English sounds and spelling.

Now we will build an LLM together by designing
an agentic system that learns the
knowledge of English syllables that you have!!!!